

Data Extraction

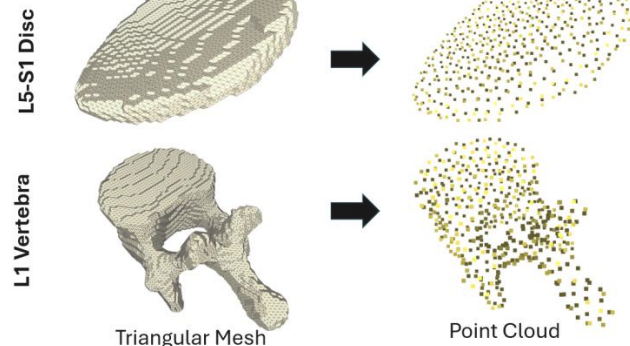
a.) Segmentation of the Spine using AI



Scan with Segmentation Mask

b.)

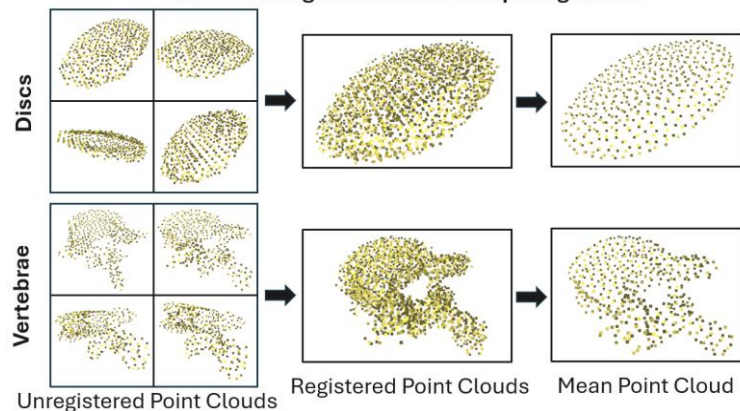
Surface Extraction and Conversion to Point Cloud



Shape Registration & Mean Shape

c.)

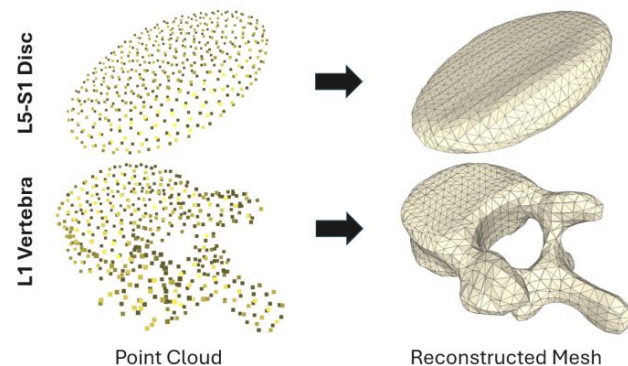
Point Cloud Registration and Computing Means



Visualization

f.)

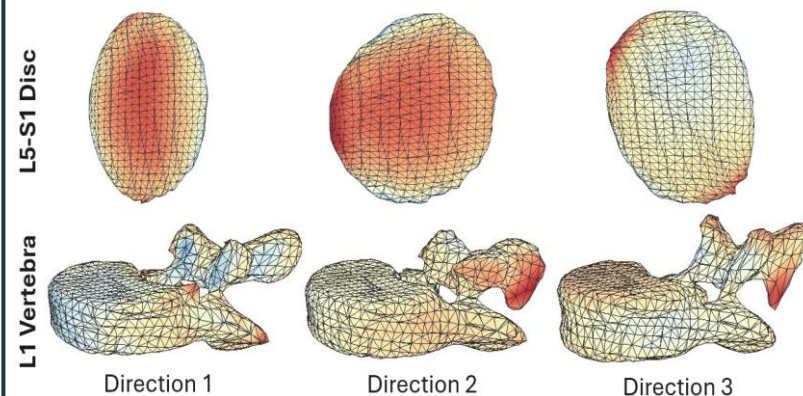
Surface Reconstruction from a Point Cloud



Principal Component Analysis (PCA)

d.)

PCA to Analyze Shape Variation from Principal Directions



e.)

Regression and Inference

Principal Component Shape Regression

